

iDeepSubMito: identification of protein submitochondrial localization with deep learning

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Abstract

Mitochondria are membrane-bound organelles containing over 1000 different proteins involved in mitochondrial function, gene expression and metabolic processes. Accurate localization of those proteins in the mitochondrial compartments is critical to their operation. A few computational methods have been developed for predicting submitochondrial localization from the protein sequences. Unfortunately, most of these computational methods focus on employing biological features or evolutionary information to extract sequence features, which greatly limits the performance of subsequent identification. Moreover, the efficiency of most computational models is still under explored, especially the deep learning feature, which is promising but requires improvement. To address these limitations, we propose a novel computational method called iDeepSubMito to predict the location of mitochondrial proteins to the submitochondrial compartments. First, we adopted a coding scheme using the ProteinELMo to model the probability distribution over the protein sequences and then represent the protein sequences as continuous vectors. Then, we proposed and implemented convolutional neural network architecture based on the bidirectional LSTM with self-attention mechanism, to effectively explore the contextual information and protein sequence semantic features. To demonstrate the effectiveness of our proposed iDeepSubMito, we performed cross-validation on two datasets containing 424 proteins and 570 proteins respectively, and consisting of four different mitochondrial compartments (matrix, inner membrane, outer membrane and intermembrane regions). Experimental results revealed that our method outperformed other computational methods. In addition, we tested iDeepSubMito on the M187, M983 and MitoCarta3.0 to further verify the efficiency of our method. Finally, the motif analysis and the interpretability analysis were conducted to reveal novel insights into subcellular biological functions of mitochondrial proteins. iDeepSubMito source code is available on GitHub at <https://github.com/houzl3416/iDeepSubMito>.

Key words: protein sequences; deep learning; protein submitochondrial localization

Introduction

Mitochondria are double-membrane bound organelles found in most eukaryotic cells and are the main sites of intracellular

oxidative phosphorylation and synthesis of adenosine triphosphate, providing chemical energy for cellular activity. In addition, mitochondria are involved in processes such as cell differentiation, cell signaling and cell apoptosis and have the ability

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to regulate the cell growth and the cell cycle [1]. From outside to the inside, mitochondria can be divided into four functional areas: the outer membrane, intermembrane space, inner membrane and matrix. This compartmentalization suggests that the proteins in each region play a specific role in the function of mitochondria. Of note, mitochondrial proteins are implicated in various human diseases including diabetes [2], myasthenia [3] and cataracts [4]. Accurate localization of proteins is crucial, therefore, for understanding their function and their role in the pathogenesis of disease.

The most direct way to obtain protein sub-cellular information is through biological experiments, such as organelle fractionation techniques with immunoblotting or labeled proteins with immunofluorescence microscopy. However, these methods are costly and time consuming, and the experimental results are highly subjective and variable. To avoid these problems, biological information and computational models have been developed to predict the localization of proteins to sub-cellular compartments[5–7].

In recent years, the accumulation of large amounts of protein sequences has enabled researchers to extract deeper structural information, facilitating the research of computer-aided subcellular protein prediction including sub-nuclear [8, 9], sub-chloroplastic [10] and submitochondrial [11–14]. Many feature extraction methods have been developed to advance this approach. Cedano et al. [15] proposed the relationship between amino acid composition and protein cell position to predict the localization of proteins. Ladunga et al. [16] identified protein domains through physicochemical attribute codes [17], while achieving protein secondary structure prediction based on a specific location scoring matrix. Chou et al. [18] raised a coding method of pseudo-amino acid composition to predict the cellular properties of proteins. In addition, several natural language model-driven approaches have also been developed to encode protein or gene sequences. Zhang et al. [19] successfully predicted CTCF-mediated chromatin loops by word2vec. The iCircRBP-DHN proposed by Yang et al. [20] improves the identification accuracy for circRNA-RBP interaction sites using the encoding method of Doc2Vec. Vang and Xie [21] performed HLA class I binding prediction on the GloVe encoding method and obtained good performances. Unfortunately, these non-contextualized approaches rely on static information that cannot fully explore the context of the sequence and the ambiguity of words.

Moreover, many learning methods have been proposed to distinguish the location of subcellular proteins. Kumar et al. [13], Nithya et al. [22] and Bian et al. [23] predicted submitochondrial protein localization based on SVM. Yu et al. [12] adopted an ensemble learning method to identify the location of submitochondrial proteins. Of note, DeepMito, using a single convolutional architecture has brought great progress in submitochondrial classification tasks [11]. Khan et al. [24] used bi-directional LSTM to identify *Plasmodium falciparum* mitochondrial proteins. Jiang et al. [25] proposed a model based on a multi-headed attention mechanism for protein subcellular and suborganelle localization prediction. Wang et al. [26] predicted submitochondrial localization based on multi-channel convolutional neural network. However, these learning methods have difficulties guaranteeing the learning of deep features and cannot adequately handle sequences with long-range dependencies. Therefore, there is room for improvement of feature-learning ability in this method.

To address this situation, we propose a novel computational method called iDeepSubMito for localizing submitochondrial

proteins, which uses the ProteinELMo to model the probability distribution over the protein sequence and then represent protein sequences as continuous vectors. In iDeepSubMito, TextCNN architecture is employed to extract local features of the sequence and then BiLSTM with attention mechanism is proposed to obtain the global context information of the submitochondrial proteins. We conducted several experiments adopting a rigorous cross-validation method without redundancy on two datasets containing the four submitochondrial compartment proteins. The experimental results revealed that the performance of iDeepSubMito is significantly better than several other deep learning methods and some traditional machine learning models. In addition, to demonstrate the generalizability of iDeepSubMito, we considered M187, M983 and MitoCarta3.0 as the independent test sets to verify the efficiency of our method. Moreover, multiple parameters and model analyses are conducted to further demonstrate the effectiveness of our model. We then analyzed the relevant biological functions of some mitochondrial proteins in motif analysis experiments and provided a website at <http://39.104.118.143:5000> to predict submitochondrial protein location online. Finally, we conducted an interpretability analysis of the training process of the iDeepSubMito network to further prove the feasibility of iDeepSubMito.

Methods

Datasets

The SM424 dataset was derived from UniprotKB/ SwissProt (release 2018_02). All non-fragmented protein sequences with experimental evidence were selected and determined subcellular localization into four submitochondrial parts (matrix, intermembrane space, outer membrane and inner membrane) [11]. After that, the CD-HIT program was used to establish identity between sequences to 40% to get non-redundant sequences. Finally, 424 mitochondrial proteins composed of 135 matrix proteins, 25 intermembrane proteins, 190 inner membrane proteins and 74 outer membrane proteins were revealed.

Similar to SM424, SM570 was derived from UniprotKB/ SwissProt by Kumar et al. [13]. The author selected full-length proteins to filter protein sequences and limited to sequences longer than 50. The sequence homology was set to 40% using the CD-HIT program. Finally, 570 protein sequences with 174 matrix proteins, 32 intermembrane proteins, 282 inner membrane proteins and 82 outer membrane proteins were obtained.

To demonstrate the generalizability of iDeepSubMito, we got protein sequence location information from the Quickgo website from a dataset containing 1050 human mitochondrial protein sequences published by Savojardo et al. [11]. Among them, 187 were annotated with GO terms equal to or descended from any of the four mitochondrial compartments: 19 proteins from the outer membrane (GO: 0005741), 70 proteins, the inner membrane (GO: 0005743), 13 proteins, the intermembrane space (GO: 0005758) and 85 proteins, the matrix (GO: 0005759). We named the dataset M187. Then, to verify the ability of model scale analysis, we chose a large-scale dataset, M983, constructed by Du et al. [27] and composed of 177 matrix, 661 inner membrane and 145 outer membrane proteins.

Feature descriptors

It can be said that the protein is the most important worker and building-block of life, which determines the function and

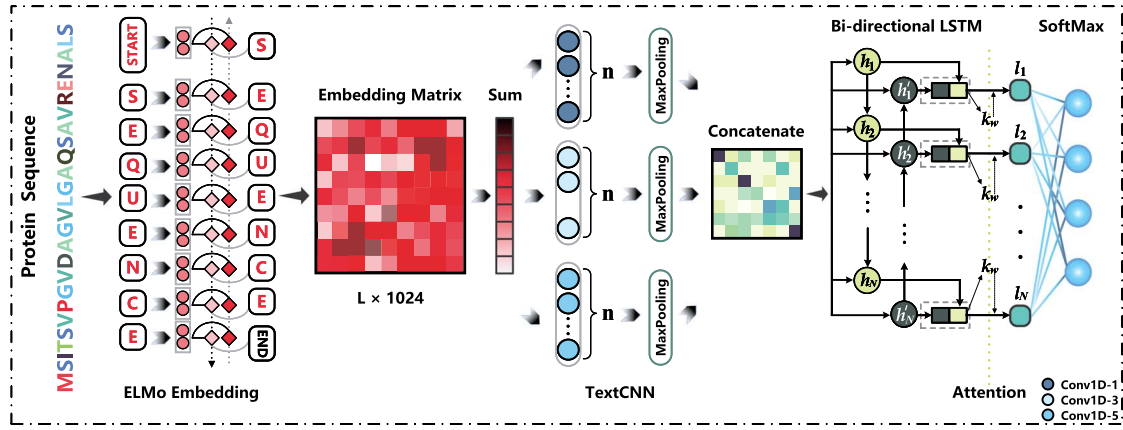


Figure 1. Overview of the proposed method, iDeepSubMito, to classify of submitochondrial proteins based on ProteinELMo embedding and deep learning methods. The protein sequence is first encoded by ProteinELMo, and then the encoded vector is passed as input to a TextCNN architecture of multiple convolutional kernels of different sizes to extract local features of the sequence. Following this, a BiLSTM with self-attention mechanism further acquires the long-range dependencies of the sequences. Finally, label classification is performed by the softmax layer.

manifestation of organisms. In order to fully explore the structural characteristics of proteins and their interconnections, inspired by SeqVec [28], we developed the coding scheme called ProteinELMo, which represents protein sequences as continuous vectors.

Unlike static word embedding methods like Word2Vec [29] or Glove [30], which divide the protein sequences into k-mer fragments that cannot fully explore the context of the sequences and the ambiguity of words, we introduce ProteinELMo to simultaneously consider the positional context information from the per-protein levels. Each amino acid has a unique code of sub-word information and sentence information in its sequence, which is essential for accurate expression of amino acid functional characteristics. ProteinELMo first maps each amino acid to a fixed-dimensional space through charCNN; we do not consider the information of adjacent residues for the time being. Then, we use BiLSTM to model the output of the charCNN layer to introduce the context of the protein sequence. The operation process of BiLSTM can be described as follows:

Given a protein sequence of N amino acids, $t_1, t_2, \dots, t_N$, the forward LSTM calculates the probability of t_k by the context-before amino acids $t_1, t_2, \dots, t_{k-1}$:

$$p(t_1, t_2, \dots, t_N) = \prod_{k=1}^N p(t_k | t_1, t_2, \dots, t_{k-1}).$$

Similar to the forward language model, backward LSTM operates in the opposite way, predicting the target amino acid by the context-after information:

$$p(t_1, t_2, \dots, t_N) = \prod_{k=1}^N p(t_k | t_{k+1}, t_{k+2}, \dots, t_N).$$

After that, a forward LM and a backward LM constitute the BiLSTM, and ProteinELMo takes the maximum log-likelihood of the language model in both directions as the objective function [31].

Indeed, in ProteinELMo, the CharCNN can learn the uncontextualized mapping of single amino acids onto a latent space

while the two layers of bidirectional LSTM (BiLSTM) can obtain the context information of surrounding residues, each protein sequence is expressed by a three-dimensional matrix ($3 * L * 1024$) where 3 represents the different embeddings generated by the three layers (one charCNN and two BiLSTM) and L is the length of the input sequence. After that, we add the vectors of the three layers to obtain a richer feature representation. The sum over this probability distribution constitutes a summarizing information for each amino acid. Then we average the 1024-dimensional embeddings of all that represents a complete protein sequence protein at the per-protein level.

The iDeepSubMito architecture

In this section, we introduce the iDeepSubMito architecture for predicting the location of mitochondrial proteins in mitochondrial compartments. The model architecture of iDeepSubMito is summarized in Figure 1 and consists of two important components: TextCNN and the BiLSTM with self-attention mechanism for exploring the contextual information and semantic features of protein sequences.

CNNs

Convolutional neural networks use an image structure to process data, have strong feature extraction ability and are applied widely in image and speech processing. A classical convolutional neural network is mainly composed of the input layer, the convolution layer, the pooling layer, the fully connected layer and the output layer.

The function of the convolutional layer is to extract features from the data in the input layer, which contains multiple convolution kernels. Each element of the convolution kernel corresponds to a weight coefficient W and a bias vector b . In the convolutional layer, the convolution kernel will regularly sweep through the input features, perform the matrix multiplication on the input features in the receptive field and superimpose the bias vector as follows:

$$v(i, j) = f((X * W)(i, j) + b) = f\left(\sum_{k=1}^n (X_k * W_k)(i, j) + b\right)$$

where k is the number of input matrices, $v(i, j)$ represents the value of the corresponding positional element of the output matrix of the convolution kernel and f is an activation, which can help the network learn complex patterns in the data. Usually, the ReLU function is used as the activation function, defined as follows:

$$f(x) = \max(0, x).$$

Finally, the output of the pooling layer is transformed into a one-dimensional vector as the input of the fully connected layer. For identification tasks, we employ the softmax function to map the vector of the hidden layer to the output layer, as follows:

$$\text{Softmax}(s_i) = \frac{e^{s_i}}{\sum_{k=1}^K e^{s_k}}$$

where s_i is the output value of the i node and K represents the number of categories of classification.

TextCNN

TextCNN is a convolutional neural network for text classification [32]. The core idea of this network architecture is to use multiple convolution kernels of different sizes to extract the key information in the protein sequences that can better obtain the local correlation between amino acids.

Let V_i be the k -dimensional word vector corresponding to the i -th amino acid in the sentence. A protein sequence of length n can be represented as follows:

$$V_{1:n} = V_1 \oplus V_2 \oplus \dots \oplus V_n$$

where $\oplus$ is the concatenation operator. Let the sliding window size of the convolution kernel be a ; the convolution layer uses a convolution kernel of $a \times k$ to extract local features of the input vector. Then, the new feature is g_i generated by $\{V_i, V_{i+1}, \dots, V_{i+a-1}\}$, which can be calculated as follows:

$$g_i = f(W \cdot V_{i:i+a-1} + b).$$

TextCNN is similar to N-gram in that it extracts local features of different widths by using multiple convolution kernels of different sizes. In Figure 1, we use convolution kernels of three different sizes for convolution. For each convolution kernel, a feature map of length $n - h + 1$ is generated after the convolution operation is completed, where n is the number of filters and h represents the kernel size. Then we extract the features through the one-max-pooling. After this, we obtain a one-dimensional vector of length n after feature extraction. Finally, we concatenate the three vectors and classify them by the softmax function.

BiLSTM with self-attention mechanism

In addition to local contextual features, long-term dependencies also exist in protein sequences. TextCNN can efficiently extract local features by convolutional kernels of different sizes; it is powerless for long-term features [20]. Therefore, in our study, we proposed to use BiLSTMs with self-attention mechanism to exploit global contextual features [33].

The hidden layer unit of LSTM is used to remember, update and utilize information by adding three gate structures, i.e. input gate (i), forget gate (f) and output gate (o). The task of the forget

gate is to accept a long-term memory M_{t-1} and decide which part of M_{t-1} to keep or abandon. For the time-step t , the forgetting factor can be calculated by the previous hidden states h_{t-1} and the current input m_t :

$$f_t = \sigma(W_f \cdot [h_{t-1}, m_t] + b_f)$$

where σ is the logistic sigmoid function. The input gate mainly controls the amount of current input m_t flowing into the memory cell, which consists of two parts. One part controls the input ratio by generating a control signal r_t , and can be defined as follows:

$$r_t = \sigma(W_r \cdot [h_{t-1}, m_t] + b_r).$$

The other part yields the candidate memory cell $\tilde{M}_t$ under the tanh layer and the control signal r_t :

$$\tilde{M}_t = \tanh(W_M \cdot [h_{t-1}, m_t] + b_M).$$

Then, the final internal memory cell can be calculated as follows:

$$M_t = f_t * M_{t-1} + r_t * \tilde{M}_t.$$

Finally, the output gate filters $\tilde{M}_t$ obtain the final output O_t by generating a control signal g_t as follows:

$$g_t = \sigma(W_g \cdot [h_{t-1}, m_t] + b_g)$$

$$O_t = g_t * \tanh(M_t).$$

It takes several time steps of information accumulation to connect distant interdependent features, and the longer the distance is, the less likely it is to be captured. In the calculation process of self-attention, the connection between any two amino acids in a protein sequence will be directly linked through a calculation step, which is conducive to the effective use of these features. The implementation of self-attention is as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where Q, K, V are the three vectors obtained by the embedding matrix. For each embedded vector, we compute a score $= Q \cdot K^T$, to keep the gradient stable, divided by $\sqrt{d_k}$ to normalize the score.

Parameter settings

The details of the parameter settings of our model and the other machine learning algorithms are described below.

Deep learning algorithms

iDeepSubMito has many hyperparameters to be optimized, and to obtain the optimal combination of parameters, we set up a list of possible values for each hyperparameter. Then, we use grid search to perform a 10-fold cross-validation experiment on each set of hyperparameters on two datasets using the average GCC index as the evaluation criteria. The test settings of combinations of different sizes convolutional kernels were $\{1, 3, 5, 7\}$,

Table 1. Optimization of the hyperparameter combinations of traditional machine learning methods

Methods	Hyperparametric combinations	Best combination
XGBoost	'n_estimators': {10, 50, 100, 300}; 'max_depth': {4, 6, 8}; 'gamma': {0, 1e-4, 1e-3, 0.1, 1}	[300, 6, 0]
GBDT	'num_iterations': {10, 50, 100, 300}; 'max_depth': {4, 6, 8}; 'num_leaves': {20, 40, 60}	[50, 6, 120]
RF	'n_estimators': {10, 50, 100, 300}; 'max_depth': {4, 6, 8}; 'min_samples_split': {10, 30, 50}	[100, 0, 2]
SVM	'kernel': {'rbf', 'linear'}; 'gamma': {0, 1e-3, 1e-4, 0.01}; 'C': {1, 20, 50, 100, 300}	[linear, 0, 1]
KNN	'n_neighbors': {1, 2, 3, 4, 5}; 'p': {0, 1, 2}; 'leaf_size': {10, 20, 30, 40, 50}	[1, 1, 40]

{7, 9, 11, 13}, {4, 5, 6, 7}, {7, 8, 9, 10}}, where each combination's number of filters were selected from 16, 32, 64, 128. In addition, iDeepSubMito uses TensorFlow 1.15 version and Keras 2.3.1 version for training, and model weight parameters in the training process use the default initialization of Keras. Moreover, when training 100 epochs a time, we add an early stopping mechanism to prevent overfitting.

For TextCNN, we also use an associated hyperparameter optimization model. The CNN architecture uses the framework proposed for DeepMito [11], which has a convolutional layer, after which the output of the convolutional layer performs global maximum pooling and global average pooling by parallelism. BiLSTM network model consists of BiLSTMs with 64 output units each. DeepForest is a depth model based on non-micro-module construction proposed by Zhou *et al.* [34], and using this, we set the number of cascade layers to 20, where each layer contains 10 estimators and each estimator includes 200 trees.

Machine learning algorithms

Firstly, we selected three ensemble learning models: Random-Forest, GBDT and XGBoost. These three algorithms are used with the scikit-learn package [35] Python 3.7. To explore other machine learning models, we examine SVM and KNN in our comparison group. Meanwhile, to conduct a fair comparison, we use the grid search model in scikit-learn [35] to find the best hyperparameters under the parameter grid as shown in Table 1.

Evaluation performance

Due to the imbalanced nature of the data, we use the Matthews correlation coefficient (MCC) as the evaluation criteria, which is a balanced measure that considers false positives and negatives.

There are four quantities (TP, TN, FP, FN) that form the confusion matrix, and the formula for each K class can be expressed as follows:

$$MCC(k) = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}.$$

Moreover, in order to evaluate the comprehensive performance of multi-classification tasks, we introduce the generalized squared correlation (GCC). Particularly, we define C_{ij} as the number of sequences belonging to the i -th class but predicted to be class j . Meanwhile, we set m_k as the sum of the proteins that truly belong to class k , while n_k represents the total number of sequences predicted to be class k . Therefore, the matrix t_{ij} is defined as follows:

$$t_{ij} = \frac{m_i n_j}{H}$$

where $H = \sum_{i=1}^K \sum_{j=1}^K C_{ij}$ is the total number of proteins.

Then we can compute the generalized squared correlation:

$$GCC = \sqrt{\frac{\sum_{i=1}^K \sum_{j=1}^K \frac{(C_{ij} - t_{ij})^2}{t_{ij}}}{H(K-1)}}$$

Results

Due to the imbalanced distribution of each category of our training dataset, we adopted 10-fold cross-validation. We divided each dataset into 10 sets of the same size by stratified sampling. During each training process, one of the subsets was the test set, one the validation set and the other eight subsets formed the training set.

Performance comparisons of different feature descriptors

To further demonstrate the effectiveness of ProteinELMo, we conduct an experiment to analyze those single different layers, their concatenation and their sum. The experimental results are summarized in Table 2. The results show that the sum operation on the embedding vectors of the three different layers is more prominent and can fuse more training information, which could lead to a richer and more accurate representation of protein features. Therefore, we selected the sum operation as the encoding method in our ProteinELMo model.

In addition, to assess the contribution of different feature-embedding approaches for submitochondrial protein localization prediction, we evaluated the performance of three feature descriptors and their combinations. In addition to ProteinELMo, the two other feature descriptors were as follows:

- Physical-chemical properties (PCP): for a protein sequence, each residue is encoded by a vector of 10 values, which represents the physical-chemical character. Overall, each sequence can be embedded by an $L \times 10$ matrix, where L is the length of the protein sequence.
- Position-specific scoring matrices (PSSM) is a feature extraction method based on evolutionary information. We obtained the PSSM through in following steps:
 1. Select a protein database as the comparison database, here, we chose Uniref90 (released June 2020) as the comparison database.
 2. Run the PSI-BLAST program by setting the iteration times to 3 and the threshold to 0.001.
 3. Finally, we obtained the PSSM in each protein encoded by a matrix with L rows and 20 columns.

The experimental results are summarized in Figure 2A. As depicted, when we considered each feature separately,

Table 2. The summary of the results for the different embedding methods of ProteinELMo

Methods	Datasets	MCC_M	MCC_T	MCC_I	MCC_O	GCC
charCNN	SM424	0.2164	0.4690	0.2664	0.0000	0.3662
LSTM1	SM424	0.6747	0.5554	0.5583	0.6144	0.6280
LSTM2	SM424	0.6166	0.6505	0.5795	0.6125	0.6558
Concatenate	SM424	0.6607	0.5752	0.5784	0.6503	0.6624
Sum	SM424	0.6825	0.6498	0.5821	0.6583	0.6803
charCNN	SM570	0.4253	0.5077	0.4731	-0.0131	0.4480
LSTM1	SM570	0.7441	0.5221	0.6293	0.5529	0.6356
LSTM2	SM570	0.6949	0.4809	0.6327	0.6644	0.6411
Concatenate	SM570	0.7336	0.4789	0.6383	0.6718	0.6411
Sum	SM570	0.7335	0.5494	0.6477	0.6995	0.6783

ProteinELMo achieves the best results, reaching 68.03% and 67.83% GCCs on both datasets, respectively, which are significantly higher than 14.00% and 17.00% for PCP, respectively, on both datasets and 53.00% and 57.00% for PSSM. To explore the performance of different combinations, we merged the two feature descriptors of PCP and PSSM. Although these results are higher than for PSSM alone, there is still a big gap with ProteinELMo's performance. In particular, for outer membrane proteins, the MCC of ProteinELMo achieved 65.83% and 69.95% on the two datasets, respectively, which was significantly higher than 47.43% and 39.97% obtained with the merged of PCP and PSSM. Then, we also compared our proposed model with the combination of ProteinELMo, PCP and PSSM. However, it did not provide better results than ProteinELMo model alone. The reason may be that the high-dimensional nature of ProteinELMo limits the expression of the evolutionary information.

In general, the performance of ProteinELMo was better than all combinations. In addition, ProteinELMo embedding is much higher than that of PSSM, and it takes about 2 min for ProteinELMo to embed protein sequences on the entire dataset.

Effect of different machine learning algorithms

In order to demonstrate iDeepSubMito's learning ability, we compared iDeepSubMito with some machine learning methods, including RandomForest (RF), GBDT, XGBoost, SVM and KNN. The experimental results obtained from the algorithms are summarized in Figure 2B. On the SM424 dataset, our proposed model outperformed all the tested machine learning algorithms, RF, GBDT, XGBoost, SVM and KNN, which produce GCCs of 45.09%, 60.08%, 63.72%, 64.05% and 53.09%, respectively, while iDeepSubMito achieves a GCC of 68.03%. On the SM570 dataset, our proposed model was superior to all other machine learning algorithms, which had GCCs of 50.83%, 60.64%, 64.34%, 64.80% and 60.99%, respectively, much lower than the 67.83% of iDeepSubMito. Of note, SVM achieved the best results of the machine learning algorithms, but poor performance in the intermembrane region, and is still not as good as that of the other deep learning algorithms. Overall, the comparative analysis demonstrated the effectiveness of our proposed model for predicting submitochondrial protein localization.

Effect of different deep learning architectures

In order to demonstrate the effectiveness of using deep neural network, we compared our proposed model with various deep learning architectures including TextCNN, CNN, BiLSTM network with self-attention mechanism (BiLSTM) and deep forest on the

two datasets. In these experiments, we used ProteinELMo as the encoding method for all of the deep learning architectures.

Figure 2C summarizes the results. Among all deep learning architectures, iDeepSubMito gave the best results, achieving 68.03% and 67.83% of GCCs on the SM424 and SM570 datasets, respectively. On the SM424 dataset, the GCC values of the four other classifiers, TextCNN, CNN, BiLSTM, and DeepForest were 67.22%, 64.93%, 67.21% and 64.75%, respectively. Indeed, we noticed that TextCNN and BiLSTM outperform CNN and DeepForest, showing the advantage of TextCNN with multiple scales of convolutional kernels for extracting local features, while BiLSTM has the ability to handle long-distance dependencies, demonstrating the effectiveness of the components of our proposed model. On SM570, the four classifiers gave 65.95%, 64.70%, 64.44% and 66.81%, respectively. Generally, merging the local feature extraction capability of TextCNN and the global contextual information acquisition capability of BiLSTM with self-attention promoted a significant improvement in identification performance. Moreover, compared to the best performing TextCNN, the MCC indices of iDeepSubMito for the identification of the outer membrane proteins were significantly improved by about 4% and 8%, respectively. Therefore, we conclude that our proposed iDeepSubMito has the ability to efficiently predict submitochondrial protein localization.

Comparing iDeepSubMito to other approaches

In this section, we compared DeepMito, which uses the deep convolutional neural network to three different feature types including residue one-hot encoding, residue PCP and evolutionary information. To conduct a fair comparison, we separated the whole dataset into 10 parts and used the average value of 10-fold cross-validations as the final result. The experimental results are depicted in Figure 2D.

For dataset SM424, iDeepSubMito provided MCC scores of 68.25%, 64.98%, 58.21% and 65.83% for matrix, intermembrane space, inner membrane and outer membrane, respectively, while in DeepMito, values were 3%, 12%, 11% and 20% lower, respectively. Overall, iDeepSubMito outperformed DeepMito by 14% GCC. On the SM570 dataset, our method achieved 73.35%, 54.94%, 64.77% and 69.95% for the four different compartments. In the outer membrane, we obtained an MCC score of 70% against 42% reported for DeepMito.

In addition, results show that our method performed relatively evenly across the four components. The variance of MCC is significantly lower than that of DeepMito, indicating that our method is better at handling unbalanced data and is

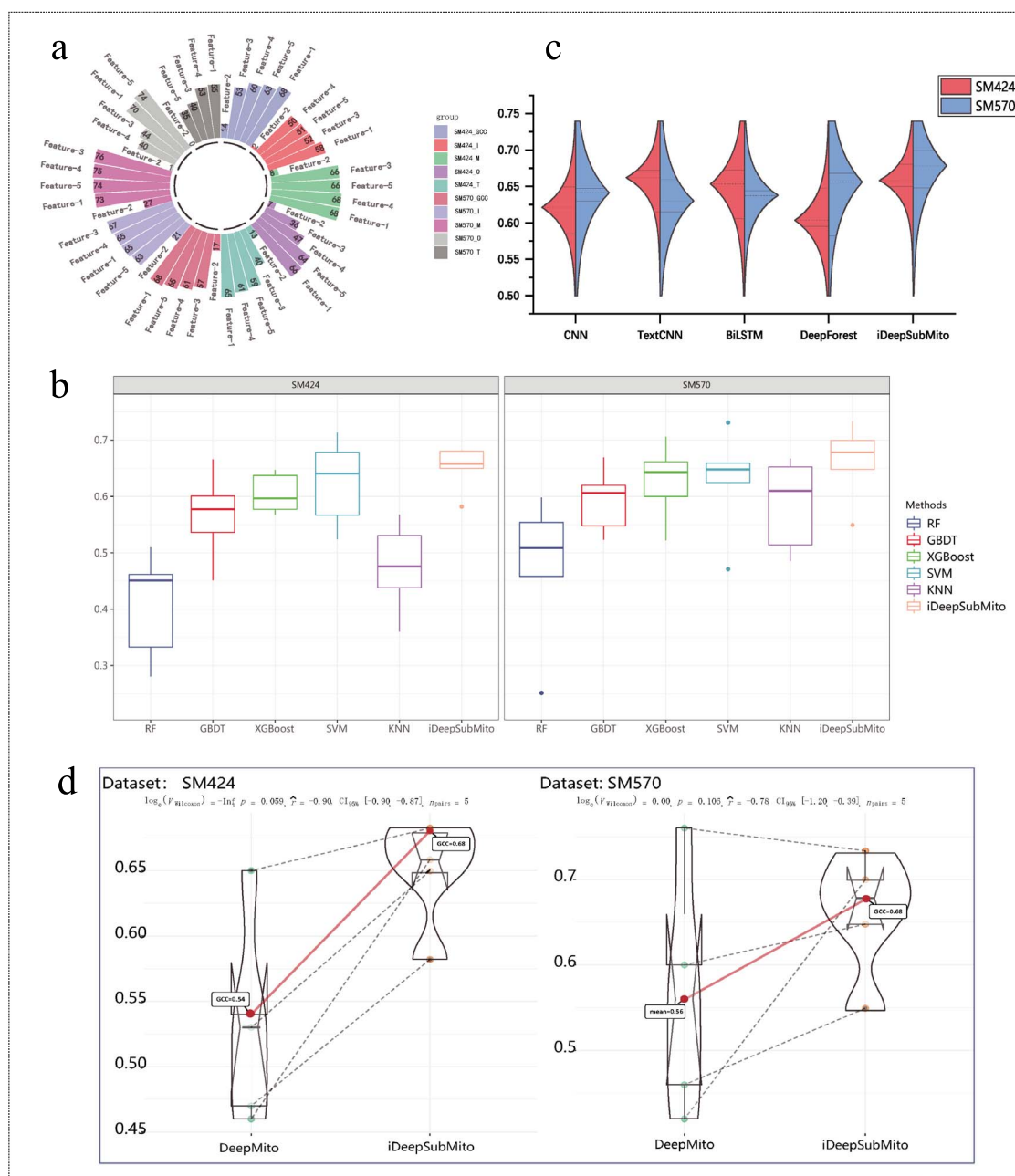


Figure 2. (A) Using the iDeepSubMito network architecture to compare the classification ability of different feature descriptors, using MCC and GCC as the evaluation criteria. The rose diagram shows the ranking results of each evaluation criteria for each of the five feature descriptors under both datasets; Feature-1, Feature-2, Feature-3, Feature-4 and Feature-5 represent ProteinELMo, PCP, PSSM, the merge of PCP and PSSM and the merge of PCP, PSSM and ProteinELMo, respectively. The result illustrates that ProteinELMo has good feature extraction ability and is more stable in the performance of different compartments. (B) The box plot evaluated the performance of the different machine learning methods. (C) Comparison of the different deep learning methods on two datasets. (D) Demonstrating the comparison between iDeepSubMito and DeepMito. The red line represents the GCC comparison and the gray line represents the MCC comparison results.

more robust. In particular, as the sample size of intermembrane proteins is only one-twentieth of the total sample, rendering its prediction accuracy was remarkably unstable, and our proposed model still yielded an MCC score of nearly 10% more than DeepMito on both datasets.

Generalization analysis

In this section, we contrasted the prediction performance of iDeepSubMito to traditional machine learning and deep learning

models including RF, GBDT, XGBoost, SVM, KNN, TextCNN, CNN, BiLSTM and DeepForest on the M187 and M983 datasets. To conduct a fair comparison, we used ProteinELMo as the only encoding method for all models except DeepMito, and we trained the prediction model on the SM424 dataset. In particular, to obtain the prediction results of DeepMito, we submitted the two datasets to the webserver published for DeepMito.

The comparison results are summarized in Figure 3A and Table 3. Overall, iDeepSubMito achieved the best predictive performance on both datasets: 68.29% GCC on the M187 dataset,

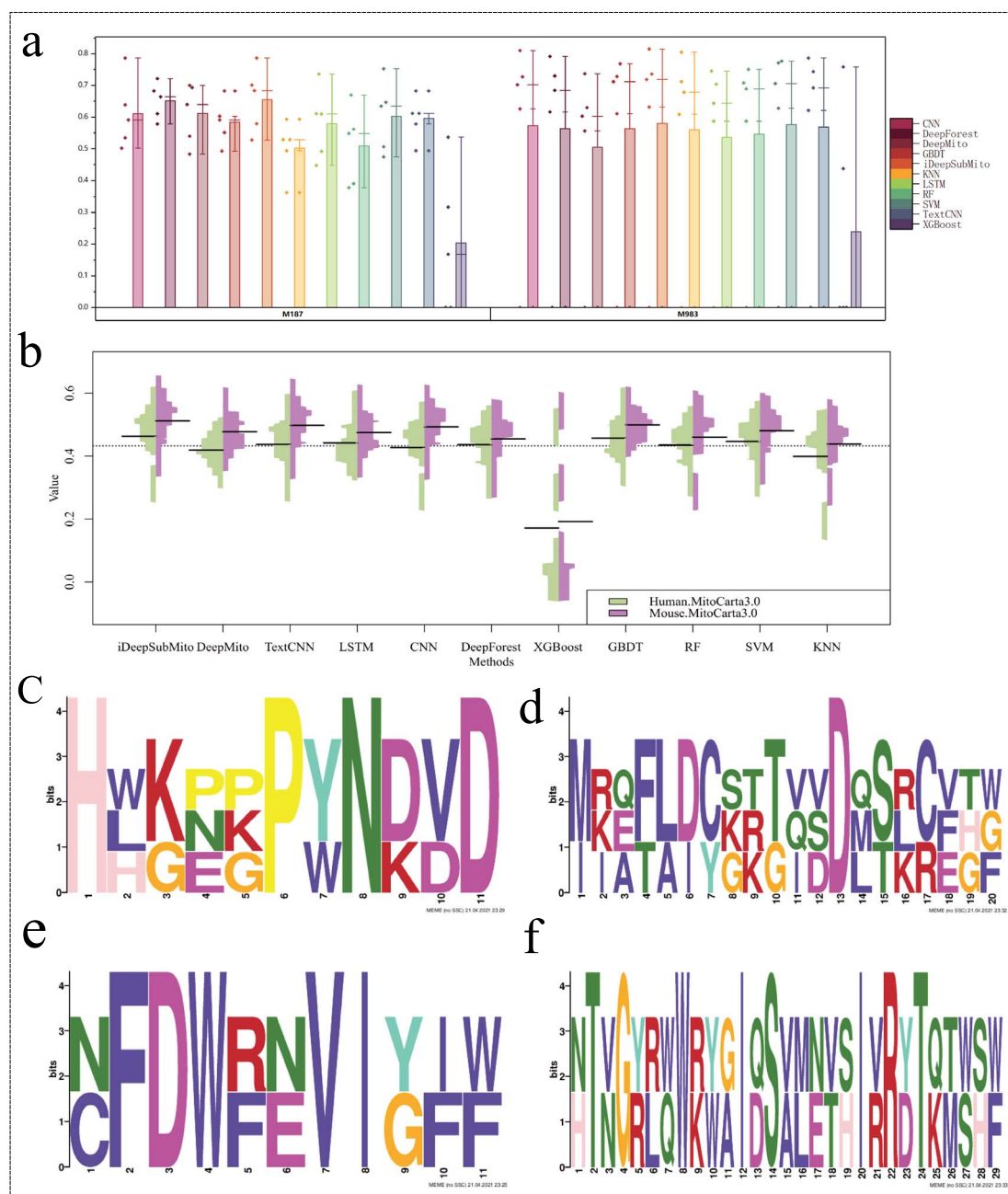


Figure 3. (A) Performance comparisons of all machine learning methods and deep learning models on submitochondrial prediction localization on M187 and M983 datasets. (B) Performance comparisons of all machine learning methods and deep learning models on submitochondrial prediction localization on MitoCarta3.0 datasets. (C, D, E, F) The motifs found for matrix, intermembrane space, inner membrane and outer membrane proteins predicted by iDeepSubMito.

significantly higher than DeepMito with 63.94%. For the deep learning algorithms, TextCNN, BiLSTM, CNN and DeepForest, the GCCs are 61.13%, 61.02%, 63.87% and 66.39%, respectively. The machine learning methods XGBoost, GBDT, RandomForest, SVM and KNN achieved 31.64%, 60.26%, 54.84%, 63.52% and 39.40%, respectively. On the M983 dataset, our algorithm had a GCC of 63.21%, which is significantly higher than DeepMito's GCC, and our algorithm outperformed DeepMito for each compartment. It is worth mentioning that most of the deep learning methods, TextCNN, BiLSTM, CNN and DeepForest, performed better than DeepMito, demonstrating the effectiveness of our proposed ProteinELMo embedding further.

iDeepSubMito showed obvious advantages in analysis of unfamiliar sequences, especially on large-scale datasets. We believe that this comes from ProteinELMo's precise feature expression and the powerful feature extraction capabilities of the iDeepSubMito network.

Comparing iDeepSubMito to MitoCarta3.0

To further validate the predictive ability of iDeepSubMito, we considered an authoritative mitochondrial protein database, MitoCarta3.0 [36], to evaluate the performance of iDeepSubMito. To generate this dataset, the organizers analyzed mitochondria

Table 3. Comparison of iDeepSubMito with other algorithms for generalization analysis results on two datasets

	iDeepSubMito	DeepMito	TextCNN	LSTM	CNN	DeepForest	XGBoost	GBDT	RF	SVM	KNN
M187											
MCC_M	0.5799	0.5394	0.5782	0.4923	0.5351	0.4297	0.5366	0.5515	0.3899	0.5064	0.3624
MCC_T	0.7864	0.6934	0.6827	0.7359	0.7864	0.7244	0.1680	0.6827	0.6695	0.7520	0.5929
MCC_I	0.5280	0.4834	0.4943	0.4479	0.5017	0.4452	0.0000	0.4934	0.3774	0.4748	0.5292
MCC_O	0.7012	0.7000	0.6118	0.6092	0.5914	0.6441	0.0000	0.5914	0.5620	0.6463	0.4943
GCC	0.6829	0.6394	0.6113	0.6102	0.6387	0.6099	0.3167	0.6026	0.5484	0.6352	0.5290
M983											
MCC_M	0.7358	0.6297	0.7433	0.7032	0.7273	0.7812	0.7585	0.7283	0.7504	0.7704	0.8049
MCC_T	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MCC_I	0.7181	0.6024	0.6924	0.6438	0.7021	0.7261	0.0000	0.7115	0.7077	0.7058	0.7113
MCC_O	0.8146	0.7366	0.7861	0.7451	0.8097	0.7281	0.0000	0.7682	0.6896	0.7769	0.6784
GCC	0.6321	0.5558	0.6215	0.5877	0.6261	0.6136	0.4379	0.6109	0.5871	0.6285	0.6083

Table 4. Comparison of iDeepSubMito with the other algorithms on MitoCarta3.0

	iDeepSubMito	DeepMito	TextCNN	LSTM	CNN	DeepForest	XGBoost	GBDT	RF	SVM	KNN
Human.MitoCarta3.0											
MCC_M	0.5617	0.4646	0.5378	0.5488	0.5145	0.5017	0.4926	0.5580	0.4801	0.5169	0.4860
MCC_T	0.3124	0.3577	0.3159	0.3831	0.2867	0.3252	0.0807	0.3651	0.3307	0.3306	0.1940
MCC_I	0.4746	0.3980	0.4443	0.4430	0.4520	0.4315	0.0000	0.4387	0.4289	0.4482	0.4890
MCC_O	0.5052	0.4522	0.4540	0.3971	0.4576	0.4860	0.0000	0.4693	0.5000	0.4926	0.4192
GCC	0.4623	0.4226	0.4350	0.4394	0.4261	0.4374	0.2849	0.4554	0.4369	0.4472	0.4063
Mouse.MitoCarta3.0											
MCC_M	0.5969	0.4822	0.5867	0.5676	0.5676	0.5183	0.5442	0.5604	0.5358	0.5415	0.5209
MCC_T	0.3949	0.4125	0.3864	0.3917	0.3845	0.3285	0.1015	0.4573	0.2873	0.3689	0.3025
MCC_I	0.4974	0.4523	0.4874	0.4538	0.4956	0.4488	0.0000	0.4775	0.4637	0.4711	0.4904
MCC_O	0.5570	0.5585	0.5326	0.4852	0.5253	0.5205	0.0000	0.5037	0.5496	0.5339	0.4306
GCC	0.5151	0.4825	0.4989	0.4775	0.4937	0.4568	0.3155	0.4991	0.4652	0.4897	0.4380

isolated from 14 tissues by mass spectrometry, assessed protein localization by large-scale GFP tagging/microscopy and then integrated the results with six other genome-scale mitochondrial localization datasets, which contain human and mouse mitochondrial protein sequences, Human.MitoCarta3.0 and Mouse.MitoCarta3.0, with corresponding submitochondrial localization information.

We did a comparison of all machine learning and deep learning models on these datasets and the results are shown in Figure 3B and Table 4. On the Human.MitoCarta3.0 dataset, we achieved a GCC of 46.21%, which is significantly higher than the performance of the other algorithms. In particular, it is almost 4% higher than DeepMito. On the Mouse.MitoCarta3.0 dataset, our method also achieved the best prediction results, reaching a GCC of 51.51%. Notably, on both datasets, our method and other deep learning methods significantly outperform DeepMito in terms of classification of matrix proteins, by nearly 10%. This implies that our ProteinELMo has a more accurate performance in feature extraction and also validates the powerful learning ability of our model.

Motif analysis

In this section, we explored the biological functions of the predicted protein sequences. First, we looked for novel, no-interval

motifs in the predicted protein sequences in each category. For the found motifs, we selected the motif with the smallest E-value and compared it to the known motif databases. Finally, we analyzed the related functions of the protein's matched motif using the MEME and TOMTOM tools in the MEME suite. The most significant motifs from the SM424 dataset can be found in Figure 3C–F.

For predicting the mitochondrial matrix proteins, we found WW domain ligands [37] (Figure 3C). Proteins containing WW domains usually control the processes of cell apoptosis and mitosis, both directly or indirectly related to some human diseases, such as Alzheimer's and Huntington's diseases. We also discovered the Pex14 ligand in the intermembrane proteins (Figure 3D); mutations in the N-terminal domain of mammalian Pex14 usually cause mitochondrial misalignment, which can induce certain diseases [38]. As shown in Figure 3E, we found the homeodomain ligand in the inner membrane proteins. The homologous box (HOX) family consists of a large number of transcription factors, which plays a crucial role in embryo development by determining the anterior/posterior and angular/caudal axis in multicellular organisms [39]. Finally, we found integrin binding sites in the outer membrane proteins [40] as shown in Figure 3F; integrins are cell surface receptors responsible for cell migration, cell-to-extracellular matrix adhesion and cell-to-cell adhesion. Besides, certain integrin proteins of mitochondria control apoptosis in the cell. Therefore,

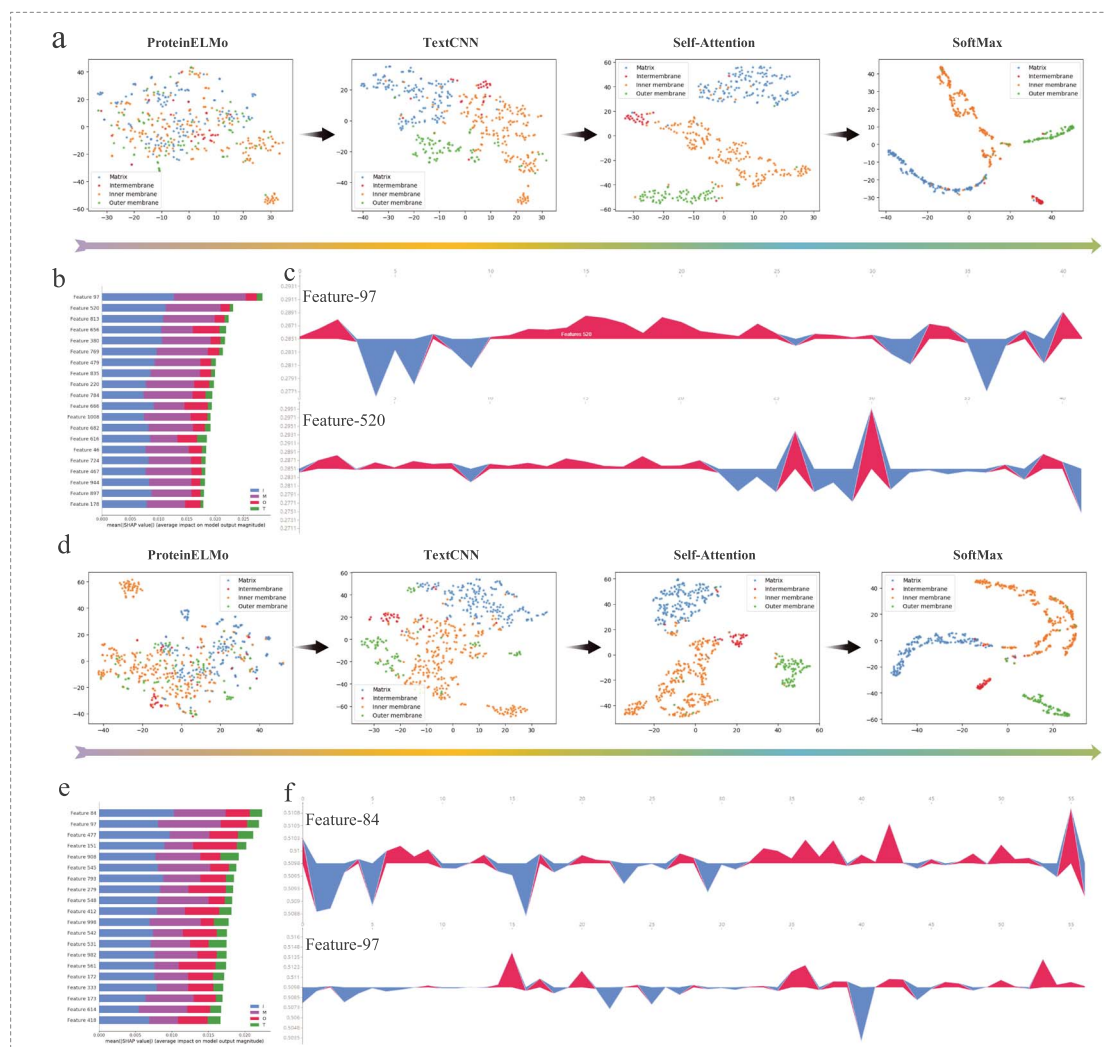


Figure 4. (A) The vector t-SNE visualization results of ProteinELMo's original embedding, the TextCNN layer, the self-attention layer and the softmax layer on the SM424 dataset, respectively. (B) The result of the top 20 features that contributed most to the classification by SHAP visualization on the SM424 dataset. (C) The influence visualization of the top two contributors on each validation set sample of SM424 for the inner membrane classification task. The red areas mean that the feature has a positive effect on the classification of the sample here, and the blue areas that it has a negative effect. (D) The visualization of t-SNE on the SM570 dataset. (E) The result of the top 20 features that contributed most to the classification by SHAP visualization on the SM570 dataset. (F) The influence visualization of the top two contributors on each validation set sample of SM570 for the inner membrane classification task. The red areas mean that the feature has a positive effect on the classification of the sample here, and the blue areas that it has a negative effect.

we conclude that our proposed model is able to discover the obvious motifs in each subcompartment, demonstrating that the ProteinELMo model can learn subtle sequence contexts from the semantics and syntax to improve performance.

Interpretability analysis

To better explain the process of model training and the embedding effect of ProteinELMo on protein sequences, we extracted the output of different intermediate layers in the process of model training on the SM424 and SM570 datasets and then projected the representation vector of each sequence into two-dimensional space using t-SNE for visualization. As shown in Figure 4A, the first figure represents the results of t-SNE for the original embedding on the SM424 dataset. Although the aggregation is not obvious, some tiny clusters can still be observed, demonstrating that the untrained ProteinELMo embedding already captured some interprotein

related information. After that, with training of TextCNN, we already saw clusters with obvious discrimination. Next, the effect was increased by the self-attention layer, in which we clearly observed the categories represented by each cluster. Finally, we used the softmax function to predict the protein sequence. The results of t-SNE visualization of the SM570 dataset are shown in Figure 4D.

Moreover, to measure the contribution of the different features, we visualized the importance of each feature using the SHAP model [41] and ranked each feature, shown in Figure 4B. We measured the top 20 features contributing the most to the training of the SM424 dataset. We can clearly see that feature-97 has the greatest impact on the training process. In addition, as depicted in Figure 4C, we evaluated the performance of the top two (Feature-97 and Feature-520) in the classification of inner membrane proteins. The red area represents the feature has a positive driving effect on the classification of the sample, which means that the larger feature value of the sample has a higher

probability of classifying it into the inner membrane. Conversely, the blue area indicates that the larger the feature value, the less likely the sample will be classified to the inner membrane. The results of SHAP visualization of the SM570 dataset are shown in Figure 4E and Figure 4F. Overall, with SHAP visualization, we not only observed which features were critical, but more importantly, the positive and negative impact of each feature was well represented.

Web server implementation

We released iDeepSubMito as a server on <http://39.104.118.143:5000>, to provide a location prediction function of submitochondrial proteins. Up to 500 sequences may be submitted as input at a time, and the predicted results will then be sent to you by e-mail. Furthermore, if you want to have the prediction results faster, or want to get the ProteinELMo embedding of the protein, you can download the relevant model and code on GitHub at <https://github.com/houzl3416/iDeepSubMito>.

Conclusion

iDeepSubMito is a more advanced submitochondrial location-prediction method, utilizing the efficient ProteinELMo encoding method and the powerful feature extraction capability of TextCNN. Its superiority was validated in multiple comparative experiments. Moreover, comparing the PCP and the PSSM containing evolutionary information, ProteinELMo embedding clearly outperforms the traditional methods in terms of coding efficiency and accuracy of feature expression. In addition, we also compared our method to DeepMito, which adopts physical-chemical attributes and PSSM to encode features and uses a simple single-layer CNN to extract local features. The experimental results showed the effectiveness of our proposed model.

Our method also performed well on independent test sets and in large-scale proteome prediction experiments. Furthermore, based on the prediction results, we also performed motif analysis to demonstrate relevant biological functions. Finally, we further investigated the strength of iDeepSubMito by conducting an interpretability analysis of the training process of the iDeepSubMito network. In summary, in the future, we will aim to apply our proposed deep model to other protein localization tasks.

Key Points

- Most protein-localizing computational methods focus on employing biological features or evolutionary information to extract sequence features, which greatly limits the performance of subsequent identification. Moreover, the efficiency of most computational models is still under explored, especially the deep learning feature, which is promising but requires improvement.
- We propose a new approach, called iDeepSubMito, using ProteinELMo as a feature extraction method and an architecture incorporating TextCNN and BiLSTM with self-attention mechanism to predict submitochondrial protein localization.
- Multiple experiments were conducted on two datasets. The experimental results indicate that iDeepSubMito significantly improves the predictive performance of submitochondrial protein localization compared to other methods. Meanwhile, iDeepSubMito has high efficiency in feature extraction

and can be used as an auxiliary tool for protein annotation.

- In addition, we tested iDeepSubMito on the human mitochondrial protein dataset as an independent test set to further verify the efficiency of our method. Finally, the motif analysis and the interpretability analysis were conducted to reveal novel insights into subcellular biological functions of mitochondrial proteins.

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References

1. McBride HM, Neuspiel M, Wasiak S. Mitochondria: more than just a powerhouse. *Curr Biol* 2006; **16**(14): R551–60.
2. Kelley DE, He J, Menshikova EV, et al. Dysfunction of mitochondria in human skeletal muscle in type 2 diabetes. *Diabetes* 2002; **51**(10): 2944–50.
3. Shichijo K, Mitsui T, Kunishige M, et al. Involvement of mitochondria in myasthenia gravis complicated with dermatomyositis and rheumatoid arthritis: a case report. *Acta Neuropathol* 2005; **109**(5): 539–42.
4. Brennan LA, Kantorow M. Mitochondrial function and redox control in the aging eye: role of msra and other repair systems in cataract and macular degenerations. *Exp Eye Res* 2009; **88**(2): 195–203.
5. Emanuelsson O, Brunak SR, Von Heijne G, et al. Locating proteins in the cell using targetp, signalp and related tools. *Nat Protoc* 2007; **2**(4): 953–971.
6. Chou KC, Shen HB. Cell-ploc: a package of web servers for predicting subcellular localization of proteins in various organisms. *Nat Protoc* 2007; **3**(2): 153–62.
7. Nanni L, Lumini A. An ensemble of support vector machines for predicting the membrane protein type directly from the amino acid sequence. *Amino Acids* 2008; **35**(3): 573–80.
8. Kumar R, Jain S, Kumari B, et al. Protein sub-nuclear localization prediction using svm and pfam domain information. *PLoS One* 2014; **9**(6): e98345.
9. Shen H-B, Chou K-C. Predicting protein subnuclear location with optimized evidence-theoretic k-nearest classifier and pseudo amino acid composition. *Biochem Biophys Res Commun* 2005; **337**(3): 752–6.
10. Savojardo C, Martelli PL, Fariselli P, Casadio R. Schloro: directing Viridiplantae proteins to six chloroplastic sub-compartments. *Bioinformatics* 2016; **33**(3): 347–353.
11. Savojardo C, Bruciaferri N, Tartari G, et al. DeepMito: accurate prediction of protein sub-mitochondrial localization using convolutional neural networks. *Bioinformatics* 2019; **36**(1): 56–64.

12. Yu B, Qiu W, Chen C, et al. SubMito-XGBoost: predicting protein submitochondrial localization by fusing multiple feature information and eXtreme gradient boosting. *Bioinformatics* 2019; **36**(4): 1074–1081.
13. Kumar R, Kumari B, Kumar M. Proteome-wide prediction and annotation of mitochondrial and sub-mitochondrial proteins by incorporating domain information. *Mitochondrion* 2018; **42**:11–22.
14. Sastri M, Darshi M, Mackey M, et al. Sub-mitochondrial localization of the genetic-tagged mitochondrial intermembrane space-bridging components mic19, mic60 and sam50. *J Cell Sci* 2017; **130**(19): 3248–3260.
15. Cedano J, Aloy P, Pérez-Pons JA, et al. Relation between amino acid composition and cellular location of proteins. *Journal of molecular biology*, 1997; **266**(3): 594–600.
16. Ladunga I. PHYSEAN: physical sequence analysis for the identification of protein domains on the basis of physical and chemical properties of amino acids. *Bioinformatics* 1999; **15**(12): 1028–1038.
17. Jones DT. Protein secondary structure prediction based on position-specific scoring matrices. *J Mol Biol* 1999; **292**(2): 195–202.
18. Chou KC. Prediction of protein cellular attributes using pseudo-amino acid composition. *Proteins* 2010; **43**(3): 246–55.
19. Zhang R, Wang Y, Yang Y, et al. Predicting CTCF-mediated chromatin loops using CTCF-MP. *Bioinformatics* 2018; **34**(13): i133–i141.
20. Yang Y, Hou Z, Ma Z, et al. iCircRBP-DHN: identification of circRNA-RBP interaction sites using deep hierarchical network. *Brief Bioinform* 2020.
21. Vang YS, Xie X. HLA class I binding prediction via convolutional neural networks. *Bioinformatics* 2017; **33**(17): 2658–2665.
22. Varadharaju, Nithya. Submitoloc: identification of mitochondrial sub cellular locations of proteins using support vector machine. *Bioinformation* 2019; **15**(12): 863–8.
23. Bian H, Guo M, Wang J. Recognition of mitochondrial proteins in plasmodium based on the tripeptide composition. *Front Cell Dev Biol* 2020; **8**:875.
24. Khan SU, Baik R. Mppif-net: identification of plasmodium falciparum parasite mitochondrial proteins using deep features with multilayer bi-directional lstm. *Processes* 2020; **8**(6): 725.
25. Jiang Y, Wang D, Yao Y, et al. A deep-learning framework for protein subcellular and suborganellar localization prediction with residue-level interpretation, 2020.
26. Wang X, Jin Y, Zhang Q. Deeppred-submito: a novel submitochondrial localization predictor based on multi-channel convolutional neural network and dataset balancing treatment. *Int J Mol Sci* 2020; **21**(16): 5710.
27. Pufeng D, Submito-pspcp YY. Predicting protein submitochondrial locations by hybridizing positional specific physicochemical properties with pseudoamino acid compositions. *Biomed Res Int* 2013; **2013**:263829.
28. Heinzinger M, Elnaggar A, Wang Y, et al. Modeling aspects of the language of life through transfer-learning protein sequences. *BMC Bioinformatics* 2019; **20**(1): 723.
29. Mikolov T, Chen K, Corrado G, et al. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*, 2013.
30. Pennington J, Socher R, Manning CD. Glove: Global vectors for word representation. *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, 2014: 1532–1543.
31. Peters ME, Neumann M, Iyyer M, et al. Deep contextualized word representations. *arXiv preprint arXiv:1802.05365*, 2018.
32. Kim Y. Convolutional neural networks for sentence classification. *arXiv preprint arXiv:1408.5882*, 2014.
33. Huang Z, Xu W, Yu K. Bidirectional LSTM-CRF models for sequence tagging. *arXiv preprint arXiv:1508.01991*, 2015.
34. Zhou, Zhi-Hua, and Ji Feng. Deep forest. *National Science Review*, 6.1 (2019): 74–86.
35. Pedregosa F, Varoquaux G, Gramfort A, et al. Scikit-learn: machine learning in python. *J Mach Learn Res* 2011; **12**(85): 2825–30.
36. Rath S, Sharma R, Gupta R, et al. MitoCarta3.0: an updated mitochondrial proteome now with sub-organelle localization and pathway annotations. *Nucleic Acids Res* 2020; **49**(D1): D1541–D1547.
37. Bork P, Sudol M. The ww domain: a signalling site in dystrophin? *Trends Biochem Sci* 1994; **19**(12): 531–3.
38. Neufeld C, Filipp FV, Simon B, et al. Structural basis for competitive interactions of pex14 with the import receptors pex5 and pex19. *EMBO J* 2009; **28**(6): 745–54.
39. Doucet-Beaupré H, Gilbert C, Profes MS, et al. Lmx1a and lmx1b regulate mitochondrial functions and survival of adult midbrain dopaminergic neurons. *Proc Natl Acad Sci* 2016; **113**(30): E4387–96.
40. Jan Y, Matter M, Pai JT, et al. A mitochondrial protein, bit1, mediates apoptosis regulated by integrins and groucho/tle corepressors. *Cell* 2004; **116**(5): 751–62.
41. Lundberg SM, Nair B, Vavilala MS, et al. Explainable machine-learning predictions for the prevention of hypoxaemia during surgery. *Nat Biomed Eng* 2018; **2**(10): 749.